Noah Weidig

GIS Analyst • Data Scientist

noah@noahweidig.com

EDUCATION

Master of Science, Forest Resources & Conservation — University of Florida — Aug 2023 – Aug 2025
Thesis: Large Wildfire Patterns in the WUI, 4.0 GPA

Bachelor of Science, Biological Sciences — Northern Kentucky University — Aug 2018 – May 2022
Ecology, Evolution, & Organismal Track, 4.0 GPA

RESEARCH & PROFESSIONAL EXPERIENCE

GIS & Remote Sensing Research Associate — University of Florida — Aug 2025
Led GIS-based analysis assessing spatial patterns and land-use impacts on community and regional risk management.

Graduate Research Assistant — University of Florida — Aug 2023 – Aug 2025
Conducted geospatial analysis to support wildland-urban interface and emergency management decision-making.

Research Assistant Intern — USDA Agricultural Research Service — May 2023 – Aug 2023
Collected, cleaned, and validated spatial datasets to support GIS-based risk assessment and land-use planning.

Fire & Recreation Intern — Student Conservation Association/US Forest Service — Jan 2023 – May 2023
Applied GIS to support spatial planning and risk assessment for land management projects.

Research Assistant — University of Cincinnati — May 2022 – Jan 2023
Conducted quantitative data analysis for biomedical research using advanced statistical modeling and visualization.

CAD Design Specialist — Garage Living — Sep 2021 – Jun 2022
Developed CAD drawings and project plans for residential remodeling projects.

Research Assistant — Northern Kentucky University — Feb 2020 – May 2022
Collected, organized, and analyzed complex ecological datasets using spatial and statistical methods.

Elections Data Specialist — Boone County Clerk's Office — Aug 2017 – Feb 2020
Managed and analyzed precinct-level GIS data for voter registration and electoral planning.

PUBLICATIONS & PRESENTATIONS

Peer-Reviewed Journal Articles

Ivey, M.A., **Weidig, N.C.**, Ivory II, A.A., Donovan, V.M. (2026). Regional differences in infilling and land-use conversion characterize woody cover increases across the Eastern United States. *Forest Ecology and Management*, *603*, 123446. doi:10.1016/j.foreco.2025.123446

Ivey, M.A., Wonkka, C.L., **Weidig, N.C.**, Donovan, V.M. (2024). Woody cover fuels large wildfire risk in the eastern US. *Geophysical Research Letters*, *51*(e2024GL110586). doi:10.1029/2024GL110586

Weidig, N.C., Wonkka, C.L., Ivey, M.A., Donovan, V.M. (2024). Changing large wildfire dynamics in the wildland–urban interface of the eastern United States. *International Journal of Wildland Fire*, *33*(12). doi:10.1071/WF24110

Weidig, N.C., Miller, A.L., Parker, A.T. (2022). The Goldilocks principle: Finding the balance between water volume and nutrients for ovipositing Culex mosquitoes (Diptera: Culicidae). *PLOS ONE*, *17*(11), e0277237. doi:10.1371/journal.pone.0277237

Theses

Weidig, N.C. (2025). *Patterns and drivers of large wildfires within and surrounding the wildland-urban interface of the eastern United States*.

Conference Papers & Presentations

Donovan, V.M., Crandall, R.M., Fill, J.M., Ivey, M.A., Ivory II, A.A., **Weidig, N.C.**, Wonkka, C.L. (2025). Unravelling changing wildfire regime dynamics in the eastern United States. *Auburn University College of Forest, Wildlife and Environment Invited Seminar*.

Ivory II, A.A., Brock, K.M., Johnson, S.A., **Weidig, N.C.**, Hallett, M.T., Acevedo, M.A., Scheffers, B.R. (2025). Southeastern Myotis (Myotis austroriparius) Use of Stormwater Sewer Systems. *Florida Bat Working Group Annual Meeting; Southeastern Bat Diversity Network*.

Ivey, M.A., **Weidig, N.C.**, Ivory II, A.A., Donovan, V.M. (2025). Woody cover increases linked to forest infilling and land conversion in the eastern United States. *International Association for Landscape Ecology - North America Annual Conference*. [link]

Alexander, H.D., Donovan, V.M., Lamounier Moura, A., Lazzaro, L.G., **Weidig, N.C.**, Ivey, M.A. (2025). Shifting forest structure and composition following decades of fire exclusion in the eastern U.S.: Implications for prescribed fire restoration and wildfires. *Proceedings of the 4th fire ecology and management symposium* (*FES-412*).

Weidig, N.C., Wonkka, C.L., Ivey, M.A., Johnson, D.J., Donovan, V.M. (2025). Fire at the fringe: Drivers of large wildfires in and near the wildland-urban interface of the eastern United States. *International Association for Landscape Ecology - North America Annual Conference*. [link]

Donovan, V.M., **Weidig, N.C.**, Ivey, M.A., Crandall, R.M., Fill, J.M., Wonkka, C.L. (2025). Unravelling changing wildfire regime dynamics in the eastern United States. *International Association for Landscape Ecology - North America Annual Conference*. [link]

Donovan, V.M., **Weidig, N.C.**, Ivey, M.A., Crandall, R.M., Fill, J.M., Wonkka, C.L. (2025). Changing large wildfire regime dynamics in the eastern United States. *Oak Woodlands & Forests Fire Consortium*.

Weidig, N.C., Wonkka, C.L., Ivey, M.A., Johnson, D.J., Donovan, V.M. (2025). Fire at the fringe: Drivers of large wildfires in and near the wildland-urban interface of the eastern United States. *Ecological Society of America*.

Weidig, N.C., Wonkka, C.L., Ivey, M.A., Johnson, D.J., Donovan, V.M. (2025). Key drivers of large wildfires in and near the eastern United States wildland-urban interface. *University of Florida Office of Graduate Professional Development Graduate Student Research Day*.

Donovan, V.M., Crandall, R.M., Fill, J.M., Ivey, M.A., **Weidig, N.C.**, Wonkka, C.L. (2024). Changing Large Wildfire Risk in the Southeastern U.S.. *Gulf of Mexico Conference*. [link]

Carl, N., Shows, C., McKeithen, J., **Weidig, N.C.**, Donovan, V.M. (2024). Rumble in the Cogongrass Jungle: Shrub Resiliency from Round 1. *Gulf of Mexico Conference*. [link]

Weidig, N.C., Wonkka, C.L., Ivey, M.A., Donovan, V.M. (2024). Large Wildfire Patterns in the Wildland-Urban Interface of the Eastern U.S.. *International Association for Landscape Ecology - North America Annual Conference*. [link]

Weidig, N.C. (2024). R Markdown: A Reproducible Workflow. *University of Florida West Florida Research and Education Center Invited Seminar*.

Weidig, N.C., Wonkka, C.L., Ivey, M.A., Donovan, V.M. (2024). Changing Large Wildfire Patterns in the Eastern United States Wildland-Urban Interface. *Northwest Florida Association of Environmental Practitioners Symposium*.

Weidig, N.C., Niang, D., Williams, M., Porema, K., Palumbo, J., Weyrich, A., Smyth, S., ... (2022). Inflammation mediated reprogramming of platelets following transcatheter aortic valve replacement. *International Society for Applied Cardiovascular Biology Biennial Meeting*.

Weidig, N.C., Miller, A.L., Parker, A.T. (2021). The Goldilocks principle: Finding the balance between water volume and nutrients for ovipositing Culex mosquitoes (Diptera: Culicidae). *Northern Kentucky University Heather Bullen Research Celebration*.

Webinars & Invited Talks

Weidig, N.C., Wonkka, C.L., Ivey, M.A., Johnson, D.J., Donovan, V.M. (2025). Burning Boundaries: Large Wildfire Patterns and Drivers in the Eastern United States Wildland-Urban Interface. *Southern Fire Exchange*. [link]

Donovan, V.M., Crandall, R.M., Fill, J.M., Ivey, M.A., **Weidig, N.C.**, Wonkka, C.L. (2025). Changing large wildfire regime dynamics in the eastern United States. *Oak Woodlands & Forests Fire Consortium*. [link]

Donovan, V.M., Crandall, R.M., Fill, J.M., Ivey, M.A., **Weidig, N.C.**, Wonkka, C.L. (2024). Increasing Large Wildfires in the Eastern United States. *Southern Fire Exchange*. [link]

Media Coverage

Georgia blaze shows how climate change has led to more wildfires in the East. *Associated Press*, 2026. [link]

Changing Large Wildfire Dynamics in the Wildland–Urban Interface of the Eastern United States: Research Brief. *Fire Lines*, 2025. [link]

Woody Cover Fuels Large Wildfire Risk in the Eastern U.S.. *Oak Woodlands & Forests Fire Consortium*, 2025. [link]

Wildfire surges in East, Southeast US fueled by new trees and shrubs. *AGU News*, 2024. [link]

AWARDS & HONORS

2026 **Outstanding Thesis in Natural Resources** University of Florida, College of Agricultural & Life Sciences
2025 Academic Year

2026 **Outstanding Thesis in Forest Resources & Conservation** University of Florida, School of Forest, Fisheries, & Geomatic Sciences
2025 Academic Year

2025 **Outstanding Teaching Assistant of the Year** University of Florida, School of Forest, Fisheries, & Geomatic Sciences
2024 Academic Year

2025 **Best Student Presentation** International Association of Landscape Ecology - North America
$300

2025 **BioLEAPS Travel Grant** International Association of Landscape Ecology - North America
$1,650

2025 **Student Travel Grant** International Association of Landscape Ecology - North America
$850

2024 **Doris and Earl Lowe and Verna Lowe Scholarship** University of Florida, College of Agricultural & Life Sciences
$2,000

2024 **IFAS Travel Grant** University of Florida, Institute of Food & Agricultural Sciences
$250

2023 **Graduate Research Assistantship** University of Florida, College of Agricultural & Life Sciences
$27,500 annually for two years

2022 **Outstanding Graduate in Biology** Northern Kentucky University, Department of Biological Sciences
$250

2018 **President's List Awardee** Northern Kentucky University
4.00 GPA · 2018–2022

2021 **1st Place, Undergraduate Student Poster Competition** Entomological Society of America
$75

2020 **Dr. Carol Swarts Milburn Research Fellowship** Northern Kentucky University
$3,500

TECHNICAL SKILLS

Research interests: Applied AI, Geospatial Analytics, Spatial Data Science, Data Visualization, Statistical Modeling, Machine Learning, Bayesian Analysis, Remote Sensing, Reproducible Research, Data Storytelling, Quantitative Ecology, Forecasting, Land Use Change, Social-Ecological Systems

Development: R, Python, GEE, ArcGIS